IS Audit & Control

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http://alphapeeler.sf.net/
reddit.com/user/alphapeeler
www.flickr.com/alphapeeler
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abdulmahmood-sss
alphapeeler@icloud.com
pinterest.com/alphapeeler

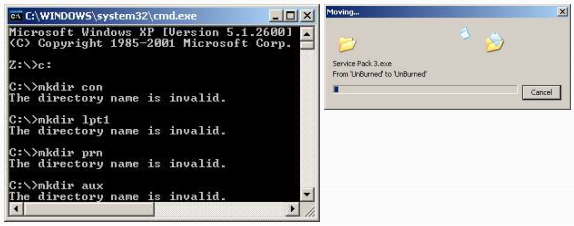
History of OS

incompatible time sharing system

Contents

- Course outline
- Is it really true? Facts!!!
- Blue screen of death – Why?
- The restart dilemma !
- POSIX
- Early Unix History
- Brief History of Linux
- Linux Timeline
- An open letter to hobbyist
- GNU / GLP
- Cygnus / Rethat
- The Cathedral and the Bazaar by Eric
- Windows Refund Day
- Homebrew Computer Club
- Open Source
- Who uses Linux
- Why openSUSE
- What is a Shell?
- Bash
- Opensource in academia

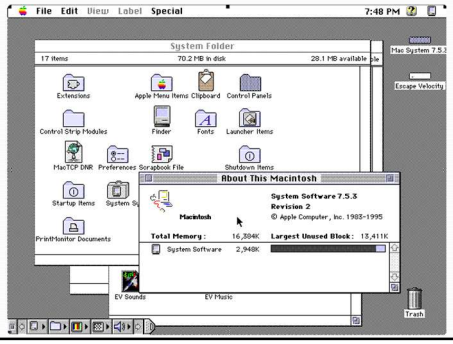
What? Is it true?



- Linux distributions don't come with disk-defragmenting utilities. But why is that?
- NTVDM ?

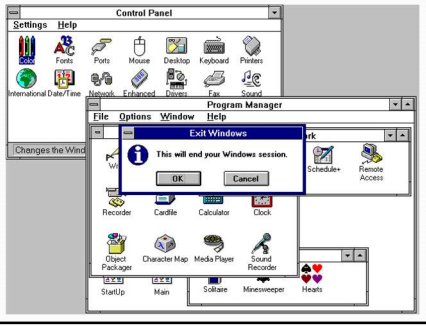
What? Is it true?

- Mac System I - 1984
- Mac System III - 1986



What? Is it true?

- Windows 3.11 or System I?
- released on April 6, 1992



What? Is it true?

Microsoft's antitrust trial, wherein the Court ruled that Microsoft Corporation's bundling of Internet Explorer with the Windows operating system was a monopolistic and illegal business practice.



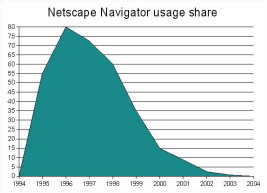
1984

Visual FoxPro
1992 - $173M
1998 - "Not a Strategic Product"



1980

Originally called QDOS, short for "quick and dirty operating system" created by Tim Paterson at Seattle Computer Co.





Forethought
1987, $14M



Visio Corp.
2000, $1.375B



Skype
2011, $8.5 B



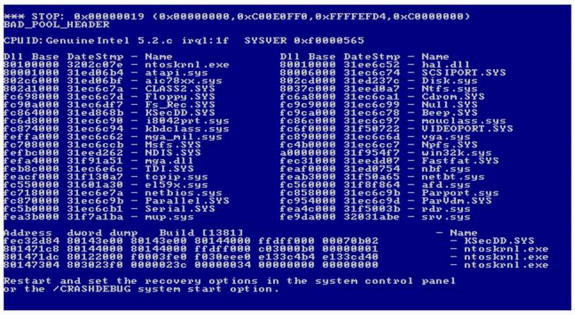
Hotmail
1997, $500 M



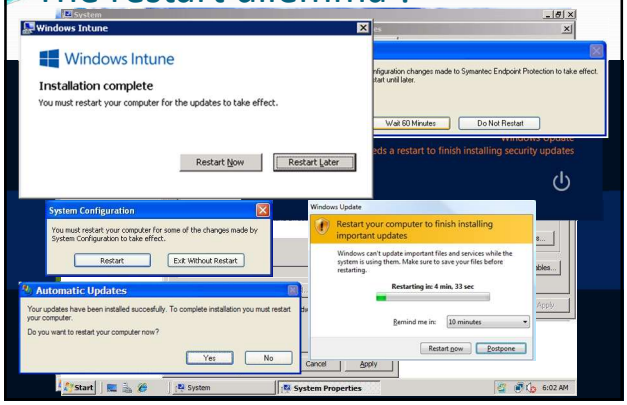
Nokia Lumia
2013, €3.79B

http://en.wikipedia.org/wiki/List_of_mergers_and_acquisitions_by_Microsoft

Blue screen of death – Why?



The restart dilemma !



POSIX

- IEEE Std 1003.1-1988, released in 1988
- Originally started as a UNIX like operating systems, but eventually grew to 19 separate documents (POSIX.1, POSIX.2 ..)
- **Fully POSIX-compliant**
  - HP-UX, Solaris, OSX
- **Mostly POSIX-compliant**
  - FreeBSD, Linux, MINIX, OpenSolaris
- **POSIX for Windows**
  - Cygwin, MinGW, Microsoft POSIX subsystem

A brief history

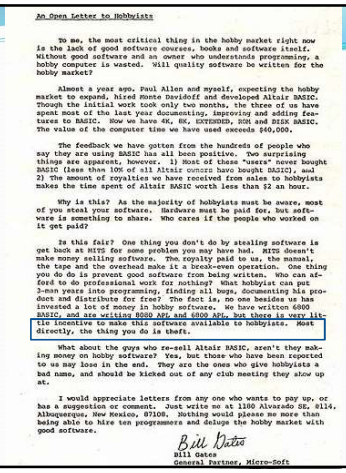
- Unix
  - Unix - multi-user, multi-tasking OS.
  - Largely hardware-independent
  - Bell Labs in **1969 by** Ken Thompson & Dennis Richie.
  - **epoch** : , Jan 1, 1970
- Linux
  - Birth of Linux : **1991** at the University of Helsinki
  - Inspired by MINIX (1987) and GNU GPL.
  - A free Unix-Like
  - 1994: version 1.0
  - Latest stable version is 3.11.4 [as of Oct 6, 2013]

Linux Timeline



Linux Timeline

- Richard Stallman, the founder of GNU Project
- Joins MIT Artificial Intelligence Lab in 1971
- The "incompatible time sharing system"
- No passwords on their computers
- What is actually a password?
- A person who enjoys a playful cleverness
- Beginning of the end of freedom.... passwords in MIT
- Mid 1970's - "Homebrew Computer Club"
- January 31, 1973 - Bill Gate wrote a open letter
- Microsoft became the pioneer of proprietary software model.
- Homebrew Computer Club - 1975



An Open Letter to Hobbyists

To me, the most critical thing in the hobby market right now is the lack of good software courses, books and software itself. Without good software and an owner who understands programming, a hobby computer is wasted. Will quality software be written for the hobby market?

Almost a year ago, Paul Allen and myself, expecting the hobby market to expand, hired Monte Davidoff and developed Altair BASIC. Though the initial work took only two months, the three of us have spent most of the last year documenting, improving and adding features to BASIC. Now we have 4K, 8K, EXTENDED, ROM and DISK BASIC. The value of the computer time we have used exceeds $40,000.

The feedback we have gotten from the hundreds of people who say they are using BASIC has all been positive. Two surprising things are apparent, however. 1) Most of these "users" never bought BASIC (less than 10% of all Altair owners have bought BASIC), and 2) The amount of royalties we have received from sales to hobbyists makes the time spent of Altair BASIC worth less than $2 an hour.

Why is this? As the majority of hobbyists must be aware, most of you steal your software. Hardware must be paid for, but software is something to share. Who cares if the people who worked on it get paid?

Is this fair? One thing you don't do by stealing software is get back at MITS for some problem you may have had. MITS doesn't make money selling software. The royalty paid to us, the manual, the tape and the overhead make it a break-even operation. One thing you do do is prevent good software from being written. Who can afford to do professional work for nothing? What hobbyist can put 3-man years into programming, finding all bugs, documenting his product and distribute for free? The fact is, no one besides us has invested a lot of money in hobby software. We have written 6800 BASIC, and are writing 8080 APL and 6800 APL, but there is very little incentive to make this software available to hobbyists. Most directly, the thing you do is theft.

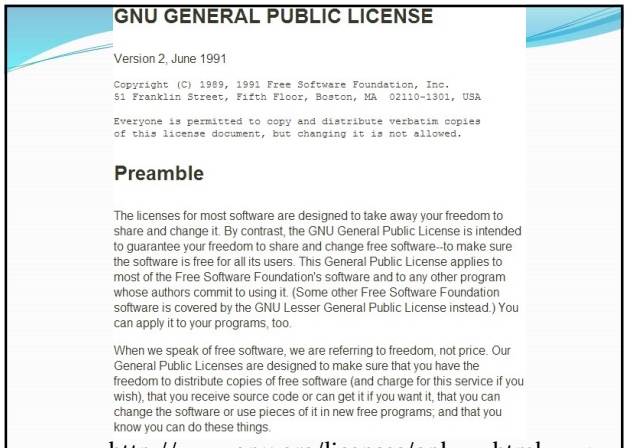
What about the guys who re-sell Altair BASIC, aren't they making money on hobby software? Yes, but those who have been reported to us may lose in the end. They are the ones who give hobbyists a bad name, and should be kicked out of any club meeting they show up at.

I would appreciate letters from any one who wants to pay up, or has a suggestion or comment. Just write me at 1180 Alvarado SE, #114, Albuquerque, New Mexico, 87108. Nothing would please me more than being able to hire ten programmers and deluge the hobby market with good software.

Bill Gates
Bill Gates
General Partner, Micro-Soft

Linux Timeline

- GNU is?
- 1980 – 1991 :
  - Richard Stallman
  - Re-writing all of the programs from scratch
- 1991 : GNU replaced practically all of the programs of UNIX.
- The concept of Copyleft
- GPL
  - Example of Copyleft
  - Linus Trovalds used this license for Linux.
  - With GPL you have the freedom of going with the people who give you good support.
  - In case of proprietary software you can only get support from only one company. Microsoft!



GNU GENERAL PUBLIC LICENSE

Version 2, June 1991

Copyright (C) 1989, 1991 Free Software Foundation, Inc.
51 Franklin Street, Fifth Floor, Boston, MA 02110-1301, USA

Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.

Preamble

The licenses for most software are designed to take away your freedom to share and change it. By contrast, the GNU General Public License is intended to guarantee your freedom to share and change free software--to make sure the software is free for all its users. This General Public License applies to most of the Free Software Foundation's software and to any other program whose authors commit to using it. (Some other Free Software Foundation software is covered by the GNU Lesser General Public License instead.) You can apply it to your programs, too.

When we speak of free software, we are referring to freedom, not price. Our General Public Licenses are designed to make sure that you have the freedom to distribute copies of free software (and charge for this service if you wish), that you receive source code or can get it if you want it, that you can change the software or use pieces of it in new free programs; and that you know you can do these things.

http://www.gnu.org/licenses/gpl-2.0.html

Linux Timeline

- Cygnus
  - Cygnus was the first business that specialized in free software
  - By the fall of 1989 – Cygnus was formed that gave support as well
  - Cygnus provided a software toolkit for free and started charging for support, but still there was no free Kernel for OS.

Linux Timeline



"I still maintain the point that designing a monolithic kernel in 1991 is a fundamental error. Be thankful you are not my student. You would not get a high grade for such a design" Prof. Andrew Tanenbaum



- **Linux** (development **:** 1991-1993)
  - **1991 :** Linux version 0.01, 10,000 LOC and 1 person
  - **1992:** Linux version 0.96, 40,000 loc and 1,000 users
  - Linux replaced Sun spark workstation[$7000]@2X speed
  - Finally people have free Unix like OS at home.
  - **1993:** Linux version 0.99, 100,000 LOC, 20,000 users
  - **1993:** Apache – the killer app of Linux
- **Red Hat - May 1995 - Mark Ewing** - 4 guys in apartment.
  - **1995:** Linux version 1.2, 250,000 LOC, 500,000 users
  - **1997:** Linux version 1.2, 800,000 LOC, 3.5 million users

Linux Timeline



- **1997**: The Cathedral and the Bazaar by Eric Raymond published this paper in a Linux conference
- **Netscape: 1998,** Frank Hacker, release s source code
- **1998:** Linux version 2.110, 1.5 million LOC, 7.5 million users

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Windows Refund Day
Monday, February 15, 1999
Foster City, California
(near Silicon Valley)

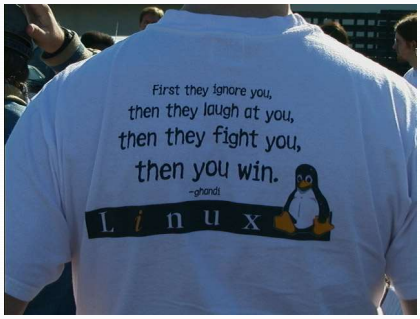
EULA

- If you do not agree to the terms of this EULA, PC manufacturer and Microsoft are unwilling to license the software product to you. In such an event … you should promptly contact PC manufacturer for instructions on a return of the unused product(s) for a refund.
- Windows 95 and a Windows 98

Windows Refund Day

- **First refund**
- Australian Geoff Bennett - refund from Toshiba Australia for the unwanted Microsoft OS included with his laptop.
- Geoff carefully read the OS's license statement, and noted that he was actually specifically required to return the software for a refund (and forbidden to use it), if he didn't consent to the license terms, which he did not.
- After a six-month epic display of mulishness, Toshiba Australia finally gave him a $110 refund.

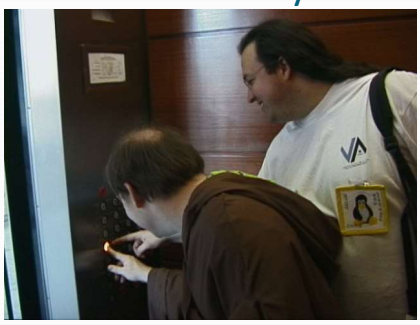
Windows Refund Day



Windows Refund Day



Windows Refund Day



Eric Raymond and Chris DiBona press the elevator button for the 9th floor. (It was locked out.)

Windows Refund Day

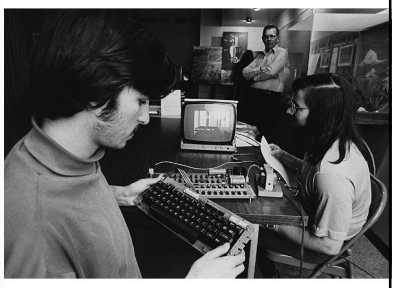
- Microsoft Planed:
  - discredit users of non-Microsoft PC OSes such as Linux and BSD.
  - without a mandatory Microsoft OS means supporting illegal software copying
  - refunds should be sought from the OEM
  - Microsoft's control of the OEMs
- **What did it achieve?**
  - change focus from OEMs to Microsoft itself.
  - Revealed Microsoft tax schemes

Homebrew Computer Club - 1975

- *Movies:*
  - *Pirates of Silicon Valley* (1999)
  - *Jobs* (2013)
- Several very high-profile hackers and computer entrepreneurs emerged from its ranks, including the founders of Apple Inc.
- Gordon French, co-founder of the Homebrew Computer Club – arranged 1st meeting - 1975

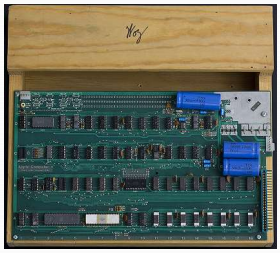
Steve Jobs and Steve Wozniak with Apple-1 computer

- Steve Jobs (left) and Steve Wozniak (right) met in a friend's garage in the late 1960s. The two of them bonded over their shared interest in electronics and practical jokes.



Homebrew Computer Club

- Steve Wozniak debuted the prototype Apple-1 at the Homebrew Computer Club in 1976. For $666.66, buyers received a simple single-board computer with 4K of RAM. A cassette-based BASIC programming language simplified interaction though users had to add a power supply, keyboard, storage system, and display to build a fully-functioning system.



Homebrew Computer Club

- **Paul Allen and Bill Gates**
- Allen and Gates had no access to an Altair when they wrote their BASIC programming language interpreter for it. They debugged the program on a DEC PDP-10 timesharing computer using a simulator of the Intel 8008 microprocessor that Allen had written.





NEWSLETTER

Homebrew Computer Club

Robert Reiling, Editor ☐ Post Office Box 626, Mountain View, CA 94042 ☐ Joel Miller, Staff Writer

Typesetting, graphics and editorial services donated by Laurel Publications, 17235 Laurel Rd., Los Gatos, CA 95030 (408) 353-3909

RANDOM DATA

By Robert Reiling

Computer clubs continue to form around the country... E. Broome would like to have material to help him get started with the "Flathead Computer Society" in the Kalispell area. His Address is P.O. Box 236, Lakeside, Montana 59922.

Did you see the SOL terminal demonstrated by Bob Marsh at the Sept. 1st meeting? An excellent design for both hobbyists and commercial users alike. It's available from Processor Technology, 6200 Hollis St., Emeryville, CA 94608. Write them for prices and specifications.

The OSI Systems Journal has been sent to all OSI customers (free--at least for the time being). It's a bi-monthly magazine with plans to go monthly in the future. There are 28 pages in the first issue (August 1976, Vol. 1, No. 1) with a hardware feature covering the OSI 440 Video Graphics System, and software, features concerning Tiny BASIC for the 6800 and a Floating Point BASIC for the 6502. Also included is a product and software catalog data. The BASIC is, of course, the 2K Tiny BASIC developed by Tom Pittman. Many of you have met Tom at the Homebrew computer Club meetings. The OSI Systems Journal is a good way to learn more about the OSI computer hardware and software along with helpful user information. The contact address is: The OSI Systems Journal, P.O. Box 134, Hiram, Ohio 44234.

KIM-1 users now have a newsletter. Eric Rehnke is producing the newsletter every 5-8 weeks, MOS Technology, Inc. is helping by providing information on all known KIM owners. The user group, however, is independent of MOS Technology, Inc. The newsletter is devoted to KIM-1 support. Subscriptions are $5.00 for the next six issues. Contact "KIM-1 User Notes," c/o Eric C. Rehnke, Apt. 207, 7656 Broadview Rd., Parma, Ohio 44134.

The RAMUG club has a new contact address. It is RAMUG, c/o Timothy O'Hare, 1211 Santa Clara Ave., Alameda, CA 94501. Write Timothy for club information. I suggest you include a stamped, self-addressed envelope.

Beware of board snatchers! Glenn Ewing reports 11 boards were taken out of his IMSAI computer. The boards are: MPU, 4 RAM-4's, SIO-2, PIO-4, PIC-8, PROM-4, IFM and PIB. Glenn suggests you consider providing good security for your computer and associated equipment. In his case the computer was in a locked office which was burglarized. In the event you have information on the above boards, write Lt. Glenn Ewing, Code 62H, Naval Post Graduate School, Monterey, CA 93940.

For family and friends of people who always wanted to know about computers, but didn't want to ask them, four easy-going classes are available starting Oct. 19th on Tuesdays from 7 to 9 p.m. You can learn how computers work and what they can and can't do. You will also have some of the jargon deciphered, see what you can do with a computer, play some games and learn to program. The cost is $25. Contact the Community Computer Center, 1919 Menalito Ave., Menlo Park, CA 94025. Phone (415) 323-1012.

A call for papers in personal computing has been issued by the 1977 National Computer Conference. The conference is scheduled for June 13-16, 1977. I have a few copies of the guidlines if you would like to submit a paper.

The First West Coast Computer Faire will be held April 16 and 17, 1977 at the San Francisco Civic Auditorium. This faire is shaping up rapidly. If you would like to lead a conference or participate in a conference session, please contact me. More information about the Faire is in the accompanying article.☐

THE FIRST WEST COAST COMPUTER FAIRE

A Call For Papers And Participation

The San Francisco Bay Area is finally going to have a major conference and exhibition exclusively concerned with personal and home computing--The First West Coast Computer Faire. The conference will be a massive one! It will take place in the Brooks Hall civic facility in Northern California: The Civic Auditorium in San Francisco. It will be a two-and-a-half day affair, starting on Friday evening and running through Sunday evening, April 15-17.

It is being sponsored by a number of local and regional hobbyist clubs, educational organizations and professional groups. These include:

•The two largest amateur computer organizations in the United States--the Homebrew Computer Club and the Southern California Computer Society.

•Much of the Bay Area chapters of the Association Of Computing Machinery--the San Francisco Chapter and the Golden Gate Chapter

•Stanford University's Electrical Engineering Department

Open Source

- Open source is free software !
- History since; 1857; 1960's
- Goes beyond hobbyists and students
- High quality software
- Open source is a viable business model
- Open source is a better software engineering methodology
- Why open source software is better:
  - Software is unlike a physical product
  - Software is heavily reuse oriented, incrementally developed

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Who uses Linux

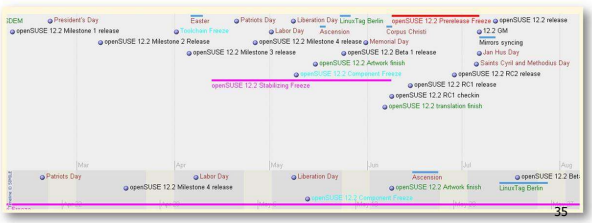
- Almost all Hackers rely on Linux
- Almost all Supercomputers runs on Linux
- Amazon.com (largest online retailer) switched entirely to Linux and saved 17 million Dollars
- Google, has a cluster of over 1000 Linux servers.
- Wikipedia
- US Department of Defense.
- French Parliament
- Commercial Bank of China
- U.S. Postal Service
- U.S. Federal Courts

Which flavor?



Why openSUSE

- Backing of big vendor - Novell
- Huge open source community
- recommended server operating systems by DELL. (Windows, RHEL, SLES)
- SUSE is the base distribution, whereas Ubuntu is extracted from Debian.
- OpenSUSE is customized for most of the development tool chains.
- Frequent build release system :



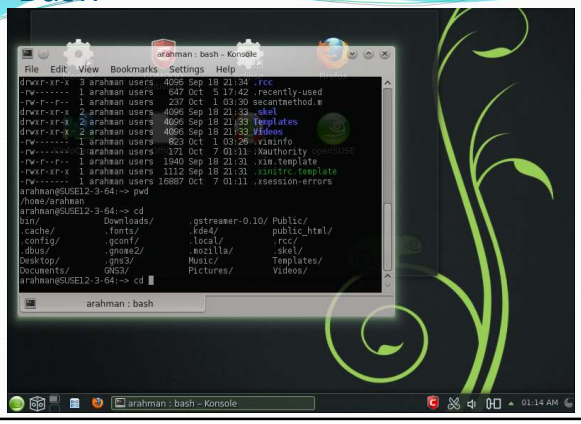
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What is a Shell?

- Just a Unix program executed when you log in
- A command interpreter
  - provides the basic user interface to UNIX utilities
- A programming language
  - program consisting of shell commands is called a **shell script**
  - you can put commands in a file and execute it:
    - First, make the file executable (**chmod u+x *script–file***)

36

Bash



37

Conclusion

- Advantages of Linux!
- Disadvantages!
- Linux does NOT hide anything!
- Need for Antivirus ? No? Why?
- Need for defragmentation tool? No!
- Only 10% Desktop computing! Why?
- Suited for Super computers
- Suited for Embedded Systems
- Very small : Tinycore (12 MB) / DSL (40MB)
- Mobile, PDA, Headsets